

# Paper 6.25 - Toolkit for Causal Code

CQE-paper-06.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

## Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

## Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-06.25.md

## Paper 6.25 - Toolkit for Causal Code

### Purpose

Paper 6.25 describes the review tools for causal code. These tools expose the typed dependency graph and its falsifiers; they do not declare the full 32-paper graph complete.

### Review Objects

The toolkit works with:

```
vertex      = paper, proof, tool, receipt, obligation, package, product
edge        = source, target, edge_type, receipt, status
closed graph = closed proof-support edges only
open edge   = tracked obligation
```

### Digital Toolkit

Primary executable files:

```
production/formal-papers/CQE-paper-06/verify_causal_code.py
production/formal-papers/CQE-paper-06/causal_code_receipt.json
```

Additional source and kernel files:

```
production/papers/CQE-paper-06/SOURCE.md
production/papers/CQE-paper-06/01-CQE-formal/FORMAL.md
production/papers/CQE-paper-06/02-CQE-tool/TOOL.md
production/papers/CQE-paper-06/03-CQE-workbook/WORKBOOK.md
production/paper-kernels/papers/CQE-paper-06/PAPER_KERNEL.md
```

The promoted verifier is the authority for Paper 6's closed claim. The kernel notes currently mark `production/papers/CQE-paper-06/02-CQE-tool/run.py` as missing, so the promoted verifier path is the replayable route for now.

### Analog Toolkit

A physical reconstruction requires:

- one card per paper/tool/receipt,
- one arrow per dependency,
- allowed edge-type labels,
- status labels,
- receipt labels.

Procedure:

```
write each dependency as an arrow
label source and target
label the edge type
attach the receipt id
mark status
trace only closed proof-support arrows
reject hidden cycles
leave open obligations open
```

## Hidden-Guess Diagnostics

When training mode is enabled, diagnostic review should ask for a prediction before revealing the receipt:

```
What fields are required on a causal edge?  
Can an open obligation be counted as proof?  
Can closed proof support contain a hidden cycle?  
Which edge carries Paper 5 into Paper 6?
```

Expected answers:

```
source, target, edge_type, receipt, status  
no  
no  
CQE-paper-05 -> CQE-paper-06, transports
```

## Boundary

Paper 6.25 is a toolkit supplement. If new edge types, statuses, or receipt rules are added, they must pass Paper 6.50's claim contract before changing the scientific paper.